MIL-A-25081 (USAF)
22 August 1955

MILITARY SPECIFICATION

AMPLIFIER, ELECTRONIC CONTROL, 2 CHANNEL, APPROACH AND NAVIGATIONAL, TYPE MB-4

1. SCOPE

1.1 This specification covers one type electronic control, 2 channel, approach and navigation amplifier, designated Type MB-4

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

NN-B-591	Boxes, Fiberboard, Wood-Cleated (For Domestic Shipment)
NN-B-621	Boxes, Wood, Nailed and Lock-Corner
NN-B-631	Boxes; Wood, Wirebound, (For Domestic Shipment)
LLL-B-631	Boxes; Fiber Corrugated (For Domestic Shipment)
LLL-B-636	Boxes; Fiber, Solid, (For Domestic Shipment)
PPP-B-601	Boxes, Wood, Cleated-Plywood

Military

MIL-E-1	Electron Tubes
JAN-P-103	Packaging and Packing For Overseas Shipment - Boxes; Wood Cleated; Solid Fiberboard
JAN-P-106	Boxes; Wood, Nailed (Overseas Type)
MIL-B-107	Boxes, Wood, Wirebound (Military Type)
JAN-P-108	Boxes, Fiberboard, Corrugated and Solid (Military Type)
MIL-P-116	Preservation, Methods of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-C-5900	Control Systems, Automatic Flight, Aircraft, General Specification For
MIL-T-9107	Test Reports; Preparation of
MIL-B-10377	Box, Wood, Cleated, Veneer, Paper Overlaid
MIL-L-10547	Liners, Case, Waterproof
MIL-E-4682	Electron Tubes, Choice and Application of

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STANDARDS

Military

MIL-STD-129	Marking For Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

405	Storage Life - Aeronautical Articles

U. S. Air Force Bulletin

67	Current List of Strategic and Critical Materials

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREPRODUCTION SAMPLE.- This specification requires the submission of preproduction samples for preproduction testing. See section 4 for details.

3.2 GENERAL SPECIFICATION.- The requirements of Specification MIL-C-5900 apply as requirements of this specification. Additional requirements shall be as specified herein.

3.2.1 REQUIREMENTS CONFLICT.- In the event the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.3 Wherever possible, strategic and critical materials, as listed in Bulletin No. 67, shall not be used.

3.4 DESIGN AND CONSTRUCTION

3.4.1 The amplifier shall operate in conjunction with the Type E-4, A-12D, or A-14 automatic pilots to produce turns and pitch attitude changes controlling the aircraft along the desired flight path as defined by right-left and up-down signals from conventional Omni-Range and Instrument Landing System radio systems.

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3.4.2 The amplifier and associated shock mount shall conform in all respects and dimensions to figure 1 and shall incorporate a single DPD-45-34P rear mounted rectangular connector identified "J-69." The mating DPD-45-33S connector shall be integral with the mount assembly. All calibration adjustments shall be accessible from the front protected by a removable cover.

3.5 ELECTRONIC PARTS.- Electronic parts and the application thereof shall be in accordance with Specification MIL-E-5400. Parts that do not appear on approved lists shall not be used unless approved by the procuring activity. Vacuum tubes of premium performance type only shall be used and shall conform to Specification MIL-E-1 and MIL-E-4682.

3.5.1 TUBE DESIGNATION.- The JAN tube designation of all electronic tubes shall be permanently marked on the chassis adjacent to the tube socket.

3.6 DIMENSIONS.- The dimensions of the amplifier shall conform to the dimensions shown on figure 1.

3.7 WEIGHT.- The weight of the amplifier and mount shall not exceed 18.8 pounds.

3.8 IDENTIFICATION OF PRODUCT.- Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 GENERAL.- Sampling, inspection, and tests of the amplifier shall be in accordance with Specification MIL-C-5900, and as specified herein.

4.1.1 For the purpose of this specification the qualification tests of Specification MIL-C-5900 shall be defined as preproduction tests.

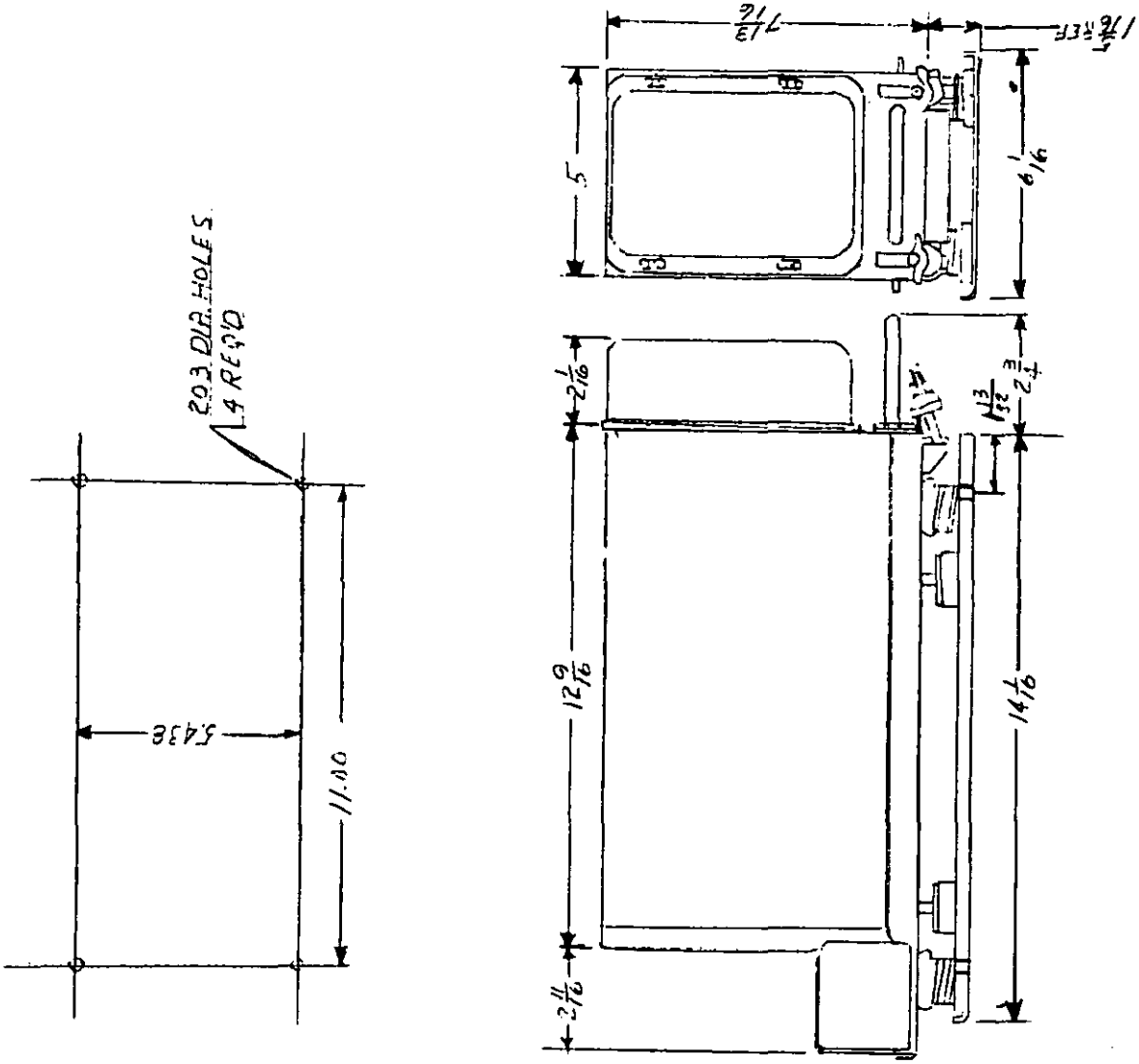
4.2 TEST CONDITIONS

4.2.1 Unless otherwise specified, all tests shall be conducted with rated voltage of 27.5 ± 0.5V dc and 115 ± 2V, 400 ± 5 cps, single phase ac.

4.2.2 SIGNAL SOURCES

4.2.2.1 Displacement signal sources shall be provided for both lateral and vertical channel inputs. These sources shall be capable of supplying and measuring to an accuracy of ±2 percent dc input signals over ranges of ±300 and ±150 mv (millivolts) into a 1000 ohm lead and shall have a maximum output impedance of 50 ohms.

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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCES:
FRACTIONS ± 1/64
DECIMALS ± .010

FIGURE 1

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These sources shall incorporate balance signal adjustments capable of supplying and measuring signals to accuracy of ±5 percent over the range of ±10 mv into the 1000 ohm load simultaneously with the application of the displacement signal. The balance signal source shall have a maximum output impedance of one ohm. Means shall be provided for removing the displacement signal while the balance signal is being adjusted.

4.2.2.2 A rate signal source shall be provided which will supply and measure to an accuracy of ±2 percent dc input signals varying uniformly at ratio of 2 and 8 mv per second into a 1000 ohm load for a period of 1 minute in each direction, i.e. uniformly from +60 to -60 mv in 60 seconds and return from -60 to +60 mv in the succeeding 60 seconds for a 2 mv per second rate signal, et cetra. The rate signal source shall have a maximum output impedance of 250 ohms.

4.2.2.3 The heading transmitter signal source shall be an AY-201S-3B synchro (or electrical equivalent) with a zero degree index reference and positive rotation reference. A dial permitting rotation from the null position to be read within 1/2° shall be provided.

4.2.2.4 The heading selector signal source shall be an AY-501-3A synchro (or electrical equivalent) with a 90° index reference and positive rotation reference. A dial permitting rotation from the null position to be read within 1/2° shall be provided.

4.2.3 POWER CONNECTIONS.- Alternating and direct current power shall be connected to the amplifier as follows:

J-69-1	Negative dc and grounded phase of the 115V, 400 cps power source. (See 6.2.1)
J-69-4	115V, 400 cps through a SPST (Single pole single throw) switch
J-69-5	+27.5V dc through a SPST switch (filaments and bias)
J-69-6	+27.5V dc through a SPST switch (to energize K-3111, lateral beam sensor) (see 6.2.2)
J-69-8	+27.5V dc through a SPST switch (lateral control)
J-69-10	+27.5V dc through a SPST switch (to energize K-3101, I.L.S.)
J-69-11	+27.5V dc through a SPST switch (to energize K-3105, flare)

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J-69-12 +27.5V dc through a SPST switch (elev. servo engaged)
J-69-16 +27.5V dc through a SPST switch (glide slope engage arming and hold)

The amplifier chassis shall be positively grounded during all tests except the insulation resistance check.

4.2.4 SIGNAL CONNECTIONS.- The various signal source connections shall be made as follows:

Heading transmitter

Stator:
"Y" - J-69-20 (blue lead for reference)
"Z" - J-69-21 (black lead for reference)
"X" - J-69-22 (yellow lead for reference)

Rotor:
"H" - 26V, 400 cps., "C" phase (red lead for reference)
"C" - 26V, 400 cps., "A" phase (ground) (black lead for reference)

Heading selector

Stator:
"X" - J-69-23 (yellow lead for reference)
"Z" - J-69-24 (black lead for reference)
"Y" - J-69-25 (blue lead for reference)

Rotor: - (Provide means for measuring this signal.)
"C" - J-69-26 (black lead for reference)
"H" - J-69-27 (red lead for reference)

Lateral input signal - J-69-28 and J-69-29 (see note)
Vertical input signal - J-69-32 and J-69-33 (see note)

NOTE: During all functional tests the lateral and vertical inputs must be connected to their signal sources or else shorted.

4.2.5 OUTPUT CONNECTIONS.- The lateral and vertical channel output voltages shall be measured with a vacuum tube voltmeter and shall be loaded as follows:

Lateral Output - J-69-30 and J-69-31; resistive load of 680 ohms ±2 percent
Vertical Output - J-69-37 and J-69-38; resistive load of 6200 ohms ±2 percent

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4.2.6 SIGNAL POLARITY DEFINITIONS

4.2.6.1 Positive input signals are defined as follows:

Lateral - J-69-29 positive with respect to J-69-28
Vertical - J-69-33 positive with respect to J-69-32
Heading - The instantaneous polarity of J-69-27 with respect to J-69-26 the same as J-69-4 with respect to J-69-1.

4.2.6.2 Positive output signals are defined as follows:

Lateral - The instantaneous polarity of J-69-31 with respect to J-69-30 the same as J-69-4 with respect to J-69-1.
Vertical - The instantaneous polarity of J-69-38 with respect to J-69-37 the same as J-69-4 with respect to J-69-1.

4.3 PREPRODUCTION TESTS.- Preproduction tests shall be performed by the contractor in accordance with the following paragraphs:

4.3.1 PREPRODUCTION TEST SAMPLES.- Preproduction test samples shall consist of the following:

4.3.1.1 When specified in the contract, three amplifiers shall be subjected by the contractor to the preproduction tests specified herein.

4.3.1.2 When specified in the contract, three amplifiers shall be submitted to the procuring activity at the time the preproduction sample test report is submitted for approval. The samples will be used by the procuring activity for:

a. A review of the mechanical construction of the amplifiers.
b. Repetition of any tests which may be deemed necessary after a review of the contractor's test report.
c. Government reference purposes.

4.3.2 PREPRODUCTION TEST REPORT.- After completion of the preproduction tests by the contractor, three copies of a complete report shall be furnished the procuring activity. The preproduction test report shall be in accordance with Specification MIL-T-9107.

4.3.3 The preproduction tests shall consist of the qualification tests specified in Specification MIL-C-5900 and the tests specified herein.

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4.4 ACCEPTANCE TESTS.- The acceptance tests shall consist of the individual tests and the sampling tests.

4.4.1 INDIVIDUAL TESTS

4.4.1.1 INSULATION RESISTANCE.- The insulation resistance between the chassis and all pins of J-69 shall be 25 megohms minimum measured with a 500V dc megger.

4.4.1.2 PRELIMINARY.- Connect the lateral displacement, vertical displacement, heading transmitter, and heading selector signal sources as specified in 4.2.4. Adjust all sources to zero or null positions. These sources shall remain connected during all tests unless otherwise specified. Set all tap switches to zero. The amplifier shall be turned on by applying the specified voltages to J-69-4 and J-69-5.

NOTE: In the following, all power and signal connections and all tap switch settings shall remain as successively specified throughout the indicated order of the tests.

4.4.1.3 LATERAL CHANNEL

4.4.1.3.1 LOCALIZER CONFIGURATION

4.4.1.3.1.1 CONTINUITY TEST.- Continuity tests shall be made between J-69-17 and J-69-18 and between J-69-16, J-69-13 and J-69-12.

4.4.1.3.1.2 LATERAL BEAM SENSOR.- Set localizer displacement tap switch to position 5 and apply 27.5V dc to J-69-10, J-69-8, and J-69-6. Apply 200 mv of lateral displacement input. The lateral output shall not exceed 0.05V. (The lateral follow-up servo or rate generator will rotate continuously.) Remove 27.5V dc from J-69-6. The lateral output shall remain unchanged until the lateral input signal is reduced to 155 ± 12 mv. When K-3111 (lateral beam sensor relay) is deenergized by decreasing the input to the specified value, the lateral output shall increase reaching 90 percent of its final output value (approximately 3.3V) in 4 ± 1.5 second.

4.4.1.3.1.3 BALANCE AND NULL OUTPUT.- Set the displacement signal source to zero, allow the lateral servo to follow up and adjust the lateral balance signal until the output meter reads a minimum value. The balance signal required to obtain this minimum shall not exceed ±8 mv. The minimum localizer output (null) shall not exceed 0.12V. Set the localizer rate tap switch to position 5. The output null shall not exceed 0.22V.

NOTE: The balance signal shall remain at the value determined above throughout the remainder of all lateral channel tests.

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4.4.1.3.1.4 DISPLACEMENT TAP SWITCH OPERATION.- Set all tap switches to zero and adjust the displacement signal source to approximately 100 mv. The output shall not exceed 0.045V. Rotate the localizer displacement tap switch from position 1 to position 5. The output voltage shall increase by approximately 40 percent at each successive tap switch position. There shall be no points of open or intermittent contact.

4.4.1.3.1.5 LOCALIZER DISPLACEMENT SENSITIVITY.- Set the localizer displacement tap switch to position 3 and with ±50 mv lateral displacement input signal the output shall be 0.44 to 0.66V. With ±150 mv input signal the output shall be 1.35 to 1.95V.

4.4.1.3.1.6 POLARITY AND PHASING OF OUTPUT.- Set the localizer displacement tap switch to position 5 and apply a positive displacement input signal sufficient to produce approximately 2.0V output. The lateral output shall have a positive polarity. The phase shift between the output voltage and the 400 cycle power supply shall not exceed ±12°. Repeat the above tests with the input signal polarity reversed.

4.4.1.3.1.7 ERECTION CUTOFF RELAY OPERATION.- Apply a positive input signal increasing from zero. The erection cutoff relay (K-3109) shall pull in between 0.5 and 1.6V of lateral output (as indicated by 27.5V dc at J-69-9). Decrease the displacement input signal slowly. The relay shall drop out above 0.3V of lateral output.

4.4.1.3.1.8 LATERAL FOLLOW-UP RANGE.- The range of the lateral follow-up servo shall be 168 ± 12 mv of input in each direction. (Beyond the limit stop range, the servo will rotate continuously.)

4.4.1.3.1.9 LOCALIZER RATE TAP SWITCH OPERATION.- Set all tap switches to zero and connect the rate input signal source to the lateral input. Adjust the rate signal to 2 mv per second. The lateral output shall not exceed 0.06V. Rotate the rate tap switch from position 1 to position 5. The lateral output shall increase by approximately 40 percent at each successive tap switch position. There shall be no points of open or intermittent contact.

4.4.1.3.1.10 LOCALIZER RATE SENSITIVITY.- Set the localizer rate tap switch to position 3 then with a rate input signal of 2 mv per second the lateral output shall be 0.56 to 0.89V. With an input of 8 mv per second the output shall be 2.20 to 3.60V.

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4.4.1.3.1.11 RATE POLARITY AND PHASING.- Apply a 2 mv per second input. The output shall have positive polarity as the rate input signal is changing in the negative to positive direction. The lateral rate output shall be in phase or 180° out of phase with the alternating current supply voltage to within ±12°.

4.4.1.3.2 APPROACH CONFIGURATION

4.4.1.3.2.1 CONTINUITY TEST.- Establish the approach configuration by applying 27.5V dc to J-69-12 and J-69-16 with zero glide path input signal. Continuity shall exist between J-69-16 and J-69-14. There shall be no continuity between J-69-16 and J-69-13.

4.4.1.3.2.2 APPROACH SENSITIVITY TAP SWITCH OPERATION.- Set the approach filter tap switch to position 3 and the approach sensitivity tap switch to position zero. Adjust the lateral displacement input signal to approximately 30 mv. The lateral output shall not exceed 0.45V. Rotate the approach displacement tap switch from position 1 to position 5. The lateral output shall increase by approximately 40 percent at each successive tap switch position. There shall be no points of open or intermittent contact.

4.4.1.3.2.3 APPROACH DISPLACEMENT SENSITIVITY.- Set the approach sensitivity tap switch to position 4 and with 50 mv input signal the output shall be 0.88 to 1.32V. With 150 mv input signal, the output shall be 1.95 to 2.85V.

4.4.1.3.2.4 APPROACH RATE TAP SWITCH OPERATION.- Short out the approach displacement output by connecting terminals 2 and 21 of front panel terminal board TB-3109 together and adjust the lateral rate input signal to 2 mv per second. Rotate the approach sensitivity tap switch from position 1 to position 5 and the lateral output voltage shall increase by approximately 40 percent between positions 2 and 3 and 3 and 4. The output at tap switch positions 1 and 5 shall be identical to the output at positions 2 and 4 respectively. There shall be no points of open or intermittent contact.

4.4.1.3.2.5 APPROACH RATE SENSITIVITY.- Set the approach filter tap switch to position 5 and the approach sensitivity tap switch to position 2. Short out the displacement component as in 4.4.1.3.2.4 above (TB-3109-2 and TB-3109-21 connected together). With a rate input signal of 2 mv per second, the output shall be 0.28 to 0.45V. With an input of 8 mv per second, the output shall be 1.11 to 1.81V.

NOTE: After this test remove the short between TB-3109-2 and TB-3109-21.

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4.4.1.3.2.6 APPROACH HOLDING FUNCTION.- Remove 27.5V dc from J-69-16 to deenergize vertical engage relay K-3107 and there shall be no continuity between J-69-16 and J-69-14. The lateral output in response to lateral input signals shall still be governed by the approach sensitivity tap switch.

4.4.1.3.3 RANGE BRACKET CONFIGURATION

NOTE: If at any time in this configuration, the position of the lateral follow-up servo corresponds to an input of 50 mv or less and simultaneously the output is less than that required to hold in the erection cut-off relay, the coupler will switch to the Range On-Course configuration. It will be necessary to momentarily re-energize lateral beam sensor relay K-3111 to reestablish the bracket configuration. The range displacement tap switch should be set to zero during all bracket tests unless otherwise specified.

4.4.1.3.3.1 ENGAGE FUNCTION.- Set the lateral displacement input signal to +200 mv and the heading transmitter and heading selector synchros to zero degrees (null position). Set the range heading tap switch to position 3 and remove 27.5V dc from J-69-10 and J-69-8. Apply 27.5V dc to J-69-6. The lateral output shall be 0.06V maximum. Apply 27.5V dc to J-69-8 and remove from J-69-6. The lateral output shall reach 90 percent of its final value (approximately 4.5V) in 4 ± 1.5 seconds.

4.4.1.3.3.2 BRACKET DISPLACEMENT TAP SWITCH OPERATION.- Reduce the lateral displacement input signal to approximately +70 mv and set the range heading tap switch to zero. The output shall be 0.04V maximum. Rotate the range heading tap switch from position 1 to position 5. The lateral output shall increase in approximately 40 percent steps at each successive tap switch position. There shall be no points of open or intermittent contact.

4.4.1.3.3.3 BRACKET DISPLACEMENT SENSITIVITY.- Set the range heading tap signal to position 2. With 75 mv lateral displacement input signal, the output shall be 1.21 to 1.82V. With 150 mv input the output shall be 2.40 to 3.60V.

4.4.1.3.3.4 HEADING TAP SWITCH OPERATION.- Set the displacement input signal to +100 mv. (Re-establish bracket configuration if necessary). Connect front panel terminals TB-3109-7 and TB-3109-20 together to short the displacement component of the lateral output and set the heading transmitter synchro to 20° then with the range heading tap switch on zero, the output shall be 0.05V maximum. Rotate the heading tap switch from position 1 to position 5 and the lateral output shall increase by approximately 40 percent at each successive tap switch position. There shall be no points of open or intermittent contact.

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4.4.1.3.3.5 HEADING SENSITIVITY.- Set the heading tap switch to position 4, apply 100 mv displacement input signal, and short the displacement component of the lateral output (as in 4.4.1.3.3.4). Apply 27.5V dc to J-69-6 and remove from J-69-8 to re-establish automatic pilot configuration and permit the integrator to follow up to its null. Apply 27.5V dc to J-69-8 and remove from J-69-6. With the heading transmitter set to 20° (heading selector synchro at zero), the lateral output shall be 1.85 to 2.95V. With the heading transmitter set to 90°, the output shall be 4.20 to 6.80V.

4.4.1.3.3.6 HEADING POLARITY AND PHASING.- Set the range tap switch to position 5. With 100 mv displacement input and the displacement component of the lateral output shorted (as above), rotate the heading transmitter counterclockwise approximately 10° from the zero position. The lateral output shall have a positive polarity. Rotate the heading selector synchro 10° clockwise from its zero position and the lateral output voltage shall increase. The phase shift between the lateral output and the 400 cycle power supply shall not exceed ±12°.

4.4.1.3.3.7 BRACKET INTEGRATOR STOP FUNCTION.- Set the range displacement and range heading tap switches to postion 3 and measure heading signal input across terminals J-69-26 and J-69-27. Remove the clip lead connection between front panel TB-3109-7 and TB-3109-20. Set the displacement input signal to zero and set the heading transmitter and heading selector synchros to zero. (This establishes the on-course configuration.) Adjust the lateral displacement input signal to 150 mv. The heading signal input (J-69-26 and J-69-27) shall increase to approximately 3V as the integrator differential synchro rotates to its limit stop. Apply 27.5V dc to J-69-6 and remove from J-69-8 (to establish the automatic pilot configuration). The heading input signal shall immediately begin to decrease as the integrator follows up. When the heading input signal reaches approximately 1.0V, apply 27.5V dc to J-69-8 and remove from J-69-6 (establishing the bracket configuration). The heading input signal shall remain constant (integrator stopped).

4.4.1.3.3.8 BRACKET OVER STATION SENSOR OPERATION.- Measure the lateral output signal (approximately 6V) and abruptly reduce the lateral input signal to 75 mv. The lateral output shall smoothly decrease in value to approximately 4V as the rate generator servo follows up to the revised input signal level. (This checks that the over-station-sensor is inoperative in the bracket configuration.)

4.4.1.3.4 RANGE ON-COURSE CONFIGURATION

4.4.1.3.4.1 ENGAGE FUNCTIONS.- Remove 27.5V dc from J-69-8 and apply to J-69-6, J-69-12, and J-69-16. Set the heading transmitter and heading selector synchros to zero degrees. Set the range heading tap switch to position 0 and the range displacement

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tap switch to position 5 and adjust the lateral displacement signal to +75 mv. Allow the rate generator to follow up the input and apply 27.5V dc to J-69-8 and remove from J-69-6. The lateral output shall be 0.08 maximum. Slowly decrease the lateral displacement input and when the input is decreased to 55 ± 20 mv the lateral output shall abruptly increase to approximately 2.0V indicating the on-course configuration has been established. Repeat the above test with the displacement input signal polarity reversed. The same limits shall apply.

4.4.1.3.4.1.1 Apply 27.5V dc to J-69-6 and remove from J-69-8. Set the range heading tap switch to position 5 and the range displacement tap switch to position 0. Adjust the lateral input signal to +35 mv and the heading transmitter and heading selector synchros to zero. Apply 27.5V dc to J-69-8 and remove from J-69-6. The lateral output shall increase to approximately 2.0V. Gradually reduce the input signal and when the erection cutoff relay K-3109 is deenergized (no continuity between J-69-8 and J-69-9) at about 0.7V lateral output, the on course configuration shall be established and the lateral output shall drop to less than 0.1V.

4.4.1.3.4.2 ON-COURSE DISPLACEMENT TAP SWITCH OPERATION.- Set the heading tap switch to position 0 and the range displacement tap switch to position 0 then adjust the lateral input signal to 15 mv. The lateral output shall be 0.06V maximum. Rotate the range displacement tap switch from position 1 to position 5 and the lateral output shall increase by approximately 40 percent at each successive tap switch position. There shall be no points of open or intermittent contact.

4.4.1.3.4.3 ON-COURSE DISPLACEMENT SENSITIVITY.- Set the range heading tap switch to position zero and the range displacement tap to position 1. With 50 mv input signal the output shall be 0.85 to 1.35V. Set the range displacement tap switch to position 3. With 150 mv input the output shall be 2.10 - 3.30V.

NOTE: If the input signal is altered rapidly, the over-station sensor may be energized. It will then be necessary to wait until the output returns to its final steady state value before continuing the test.

4.4.1.3.4.4 INTEGRATOR TAP SWITCH OPERATION.- Adjust the lateral input signal to +50 mv and set the range displacement tap switch to zero. Read the integrator tap switch voltage between front panel terminals TB-3109-16 and TB-3109-7. The tap switch voltage shall be zero. Rotate the range displacement tap switch from position 1 to position 5 and the integrator tap switch voltage shall increase approximately 40 percent for each successive tap switch position. There shall be no points of open or intermittent contact.

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4.4.1.3.4.5 Set the range displacement tap switch to position 3 and measure the heading input signal (J-69-26 and J-69-27). Adjust the lateral displacement input signal to -50 mv and allow the integrator to drive through zero to about 0.25V of heading input error signal. Reverse the lateral displacement input signal and measure the heading input signal for 100 seconds starting as the signal goes through zero. The heading error input signal shall be 0.83 to 1.32V. Repeat the test for the opposite polarities. The same limits shall apply and the readings in the two directions shall not differ by more than 0.15V.

4.4.1.3.4.6 DIRECTION OF INTEGRATOR ROTATION.- Set the range heading tap switch to position 2 and the range displacement tap switch to position 2. Apply 27.5V dc to J-69-6 and remove from J-69-8 to allow the integrator to follow up to its null position. Adjust the lateral displacement input signal to +35 mv apply 27.5V dc to J-69-8 and remove from J-69-6. After the initial "easy-engage" build up of lateral output signal, (to approximately 1.2V) the output signal shall continue to increase steadily as the integrator rotates.

4.4.1.3.4.7 OVER-STATION SENSOR FUNCTION.- Set the range heading tap switch to position 0 and the range displacement tap switch to position 1 then read the lateral follow-up servo generator voltage between front panel terminals TB-3109-8 and TB-3109-7. Adjust the lateral displacement input signal to +150 mv. The lateral output shall be approximately 3.0V. Rapidly increase the input signal to a point where the generator voltage is 2.0V. The output shall drop to less than 0.25V with a time delay of approximately 4 seconds. Rapidly reduce the input signal to 150 mv. The lateral output shall return to 90 percent of its initial full value in approximately 6 seconds.

4.4.1.4 VERTICAL CHANNEL

4.4.1.4.1 GLIDE SLOPE CONFIGURATION

4.4.1.4.1.1 ENGAGE FUNCTIONS.- Adjust the vertical displacement input signal to +50 mv and the lateral displacement input to zero then apply 27.5V to J-69-6, J-69-8, J-69-10, J-69-12 and J-69-16. Set the glide slope integrator tap switch to position 0 and the glide slope sensitivity tap switch to position 5 then remove 27.5V from J-69-6. There shall be continuity between J-69-16 and J-69-13. The vertical output shall be zero. Slowly reduce the vertical input signal and when the input signal reaches 15 ± 15 mv, relays K-3103, K-3104, K-3106, and K-3107 shall pull in establishing the glide slope configuration. Continuity shall exist between J-69-16 and J-69-14. The vertical output shall change in response to vertical input signals.

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4.4.1.4.1.1.1 Remove 27.5V dc from J-69-8, J-69-12 and J-69-16, then apply 27.5V dc to J-69-6 and adjust the vertical displacement input to -150 mv. Apply 27.5V dc to J-69-8, and J-69-12 and remove from J-69-6. There shall be continuity between J-69-12 and J-69-13 and zero vertical channel output. Apply 27.5V dc to J-69-16 and the glide slope configuration shall be engaged immediately as indicated by about 0.6V vertical output.

4.4.1.4.1.2 BALANCE AND NULL.- With the tap switches as above set the displacement input to zero and adjust the vertical balance signal until the vertical output reads a minimum. The balance signal required to obtain this minimum shall not exceed ±7 mv. The vertical output voltage null shall not exceed 0.06V. The vertical balance signal shall remain at the value determined above for the remainder of the vertical channel tests.

4.4.1.4.1.3 GLIDE SLOPE SENSITIVITY TAP SWITCH OPERATION.- Set the glide slope sensitivity tap switch to position 0 and adjust the vertical displacement input to approximately 50 mv. The vertical channel output shall be 0.035V maximum. Rotate the glide slope sensitivity tap switch from position 1 to position 5 and at each successive position the vertical output shall exhibit a rate "kick" approximately double the steady state displacement output which shall increase approximately 40 percent. There shall be no points of open or intermittent contact.

4.4.1.4.1.4 GLIDE SLOPE SENSITIVITY.- Set the glide slope sensitivity tap switch to position 3 and with 50 mv vertical displacement input signal, the output shall be 0.19 to 0.29V. With 150 mv input signal, the output shall be 0.47 to 0.71V.

4.4.1.4.1.5 POLARITY AND PHASING.- Set the glide slope sensitivity tap switch to position 5 and apply a positive vertical displacement input signal sufficient to produce approximately 0.4V output. The vertical output shall have a positive polarity and the phase shift between the output voltage and the 400 cycle power supply shall not exceed ±12°. Repeat this test with the input polarity reversed.

4.4.1.4.1.6 INTEGRATOR BALANCE.- Set the glide slope sensitivity tap switch to position 0 and the glide slope integrator tap switch to position 1. Adjust the vertical input signal until the integrator stops (vertical output remains constant). The input signal required to stop the integrator shall not exceed ±6 mv.

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4.4.1.4.1.7 INTEGRATOR TAP SWITCH OPERATION.- Set the glide slope sensitivity and glide slope integrator tap switches to position zero and apply 100 mv vertical displacement input. The glide slope integrator tap switch signal measured between front panel terminals TB-3109-7 and TB-3109-13 shall be zero. Rotate the glide slope integrator tap switch from positions 1 to 5 and the voltage between TB-3109-7 and TB-3109-13 shall increase approximately 40 percent at each successive tap switch position. There shall be no points of open or intermittent contact.

4.4.1.4.1.8 INTEGRATOR SENSITIVITY.- Set the glide slope sensitivity tap switch to position 0 and the glide slope integrator tap switch to position 3 then record the elapsed time to seconds for the vertical output to change 2.0V (1.0V each side of zero). For 150 mv input signal value the elapsed time for a 2.0V change in output shall be 32 to 50 seconds.

4.4.1.4.1.9 INTEGRATOR LIMIT.- Set the glide slope integrator tap switch to position 5 and adjust the vertical displacement input signal to ±150 mv. The vertical output shall increase to a limited value of 1.16 to 1.74V. Repeat the test for the opposite input signal polarity. The same limits shall apply.

4.4.1.4.1.10 INTEGRATOR POLARITY.- Adjust the vertical displacement input signal so as to drive the integrator to its null output position. Set the glide slope displacement tap to position 2 and apply a 50 mv vertical displacement input signal. The integrator shall produce a displacement signal which adds to the displacement component of the vertical output.

4.4.1.4.1.11 INTEGRATOR STOP AND FOLLOW-UP FUNCTIONS.- Set the glide slope sensitivity and glide slope integrator tap switches to position 3. Apply a vertical displacement input of +50 mv. When the vertical output (displacement plus integrator components) add to approximately 0.5V, remove the 27.5V dc supply from J-69-16. The displacement component of vertical output shall be shorted out immediately and only the integrator component (approximately 0.3V) shall remain. This output shall remain constant regardless of vertical input signal level. There shall be continuity between J-69-16 and J-69-13. When the 27.5V dc supply to J-69-8 is removed the vertical output shall gradually decrease towards zero as the integrator follows up. The vertical output shall not exceed 0.15V when the integrator hold relay K-3106 drops out shorting the vertical output to zero.

4.4.2 SAMPLING TEST

4.4.2.1 LOW TEMPERATURE.- After exposure to low temperature the amplifier shall be turned on and allowed to operate for a period of at least 15 minutes before any tests are made.

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4.4.2.1.1 LOCALIZER CONFIGURATION

4.4.2.1.1.1 LOCALIZER DISPLACEMENT SENSITIVITY.- Repeat the test specified in 4.4.1.3.1.5. With ±50 mv lateral displacement input, the output shall be 0.42 to 0.68V. With ±150 mv input the output shall be 1.25 to 2.05V.

4.4.2.1.1.2 LOCALIZER RATE SENSITIVITY.- Repeat the test specified in 4.4.1.3.1.10. With a rate input signal of 2 mv per second the output shall be 0.65 to 1.15V. With a rate input signal of 8 mv per second the output shall be 2.50 to 4.50V.

4.4.2.1.2 RANGE CONFIGURATION

4.4.2.1.2.1 HEADING SENSITIVITY.- With the range displacement tap switch set to position 0, the heading tap switch set to position 2, displacement input set to zero, heading transmitter and heading selector synchros on zero and the integrator followed up to its null position, establish the range on-course configuration. Set the heading transmitter synchro to 20°. The output shall be 0.85 to 1.55V.

4.4.2.1.2.2 INTEGRATOR SENSITIVITY.- Repeat the test specified in 4.4.1.3.4.5. With 50 mv input signal the heading input signal shall change from zero to 0.60 to 1.04V in 100 seconds.

4.4.2.1.3 APPROACH CONFIGURATION

4.4.2.1.3.1 GLIDE SLOPE SENSITIVITY.- Repeat the test specified in 4.4.1.4.1.4. With 50 mv displacement input signal, the output shall be 0.18 to 0.30V. With 150 mv input signal the output shall be 0.44 to 0.74V.

4.4.2.1.3.2 GLIDE SLOPE INTEGRATOR SENSITIVITY.- Repeat the test specified in 4.4.1.4.1.8. With 150 mv input, the elapsed time for a 2.0V change in output shall be 39 to 63 seconds.

5. PREPARATION FOR DELIVERY

5.1 GENERAL.- The requirements specified herein apply only to direct purchases by, or direct shipments to the Government.

5.2 PRESERVATION AND PACKAGING

5.2.1 DOMESTIC SHIPMENT.- (Immediate use without storage). Amplifiers shall be afforded preservation and packaging in accordance with the manufacturer's replacement-part practice.

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5.2.2 DOMESTIC SHIPMENT.- (Limited storage). Amplifiers shall be afforded protection against corrosion and physical and mechanical damage by application of methods used by the manufacturer to afford such protection during the process of manufacture, assembly, storage and shipment on a commercial basis. They shall be individually wrapped, bagged, boxed, or otherwise protected in a manner that will prevent the entrance of dirt, dust, or other foreign matter or direct entry of free water. Any packaging material used in direct contact with surfaces adversely affected by hygroscopic material shall be neutral within the pH range of 6.0 through 8.0.

5.2.3 OVERSEAS SHIPMENT OR EXTENDED STORAGE.- Amplifiers shall be packaged Method IIb, in accordance with Specification MIL-P-116.

5.3 PACKING

5.3.1 DOMESTIC SHIPMENT.- (Immediate use without storage). Amplifiers packaged as specified in 5.2.1 shall be packed in containers of the type, size, and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination at the lowest applicable rate. Shipping containers shall comply with the Consolidated Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

5.3.2 DOMESTIC SHIPMENT AND STORAGE.- Amplifiers packaged as specified in 5.2.2 shall be packed in wood-cleated fiberboard, nailed wood, wirebound wood, corrugated or solid fiberboard, cleated plywood, or cleated wood boxes confirming to Specifications NN-B-591, NN-B-621, NN-B-631, LLL-B-631, LLL-B-636, PPP-B-601, and MIL-B-10377, respectively. Closures shall be made in accordance with the applicable box specification. Fiberboard boxes shall conform to the special requirements of the applicable box specification. Fiberboard having a minimum dry bursting strength (Mullen test) of less than 275 pounds; shall be used. The gross weight of boxes shall not exceed 500 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.3.3 OVERSEAS SHIPMENT.- Amplifiers packaged as specified in 5.2.3 shall be packed in wood-cleated fiberboard, nailed wood, wirebound wood, corrugated or solid fiberboard, cleated wood, or cleated plywood boxes conforming to Specifications JAN-P-103, JAN-P-106, MIL-B-107, JAN-P-108, MIL-B-10377, MIL-L-10547, and PPP-B-601, respectively. The gross weight of the boxes shall not exceed 200 pounds.

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5.4 MARKING OF SHIPMENTS.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be as follows:

Amplifier, Electronic Control, 2 Channel, Approach and Navigational, Type MB-4

5.4.1 REINSPECTION DATE.- Reinspection date markings shall be in accordance with ANA Bulletin 405.

6. NOTES

6.1 INTENDED USE.- The Type MB-4 amplifier covered by this specification is intended for use with automatic pilots in making automatic approaches under instrument conditions and for automatic flight on omni-range radio courses.

6.2 DEFINITIONS

6.2.1 Wherever the term "J-69-1", or any other number combination preceded by "J" is used, it will be interpreted as "amplifier connector J-69-1".

6.2.2 Wherever the term "K-3111", or any other number combination preceded by "K" is used, it will be interpreted as "Relay K-3111".

6.3 ORDERING DATA.- Invitations for bids, contracts, and purchase orders should state the conditions for the following:

6.3.1 The method of packaging and packing required.

6.3.2 PREPRODUCTION TESTS.- It is expected that the contract or purchase order will specify that a minimum of three amplifiers will be required as preproduction samples to be subjected to preproduction tests by the contractor to determine compliance with the requirements of this specification and that three amplifiers will be required as preproduction samples to be submitted to the procuring activity.

PATENT NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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